Probabilistic Combinatorics

Sheet 1

April 22, 2024

Problem 1. Let $R(\ell, k)$ be the Ramsey number as it was defined the lecture.

1. Show that if $\ell, k \geq 2$, then $R(\ell, k) \leq R(\ell - 1, k) + R(\ell, k - 1)$.
2. Show that $R(\ell, k) < \infty$ for $\ell, k \geq 1$.

Proof.
Let $m = R(\ell, k)$. For easier intepreration, let ℓ and k be the number of vertices of induced subgraphs of K_m that are monochromatic where all of their edges are coloured red and blue, respectively.

1. Let $n = R(\ell - 1, k) + R(\ell, k - 1)$. We want to show that K_n has either an induced subgraph of K_n on ℓ vertices in which all edges are coloured red or an induced subgraph of K_n on k vertices in which all edges are coloured blue. Next, fix a vertex $v \in K_n$, and let

   - A be the set of all vertices in K_n that are adjacent to v via a red edge.
   - B be the set of all vectices in K_n that are adjacent to v via a blue edge.

   It is obvious that

   $$|A| + |B| = R(\ell - 1, k) + R(\ell, k - 1) - 1 \tag{1.1}$$

   Therefore, either $|A| \geq R(\ell - 1, k)$ or $|B| \geq R(k, \ell - 1)$. Otherwise, we have the left-hand side of the equation (1.1) less than the right-hand side of the equation (1.1).

   We firstly consider the case where $|A| \geq R(\ell - 1, k)$. First of all, it is obvious that the induced subgraph of K_n on vertices in A, which is denoted by G_A, must have one of the following properties

   - An induced subgraph of G_A on $\ell - 1$ vertices in which all edges are coloured red. Recall that the vertices in A are adjacent to v via a red edge. Therefore, the induced subgraph of K_n on vertices in $A \cup \{v\}$ must be an induced subgraph of K_n on ℓ vertices in which all edges are coloured red.
   - An induced subgraph of G_A on k vertices in which all edges are coloured blue. This is also an induced subgraph of K_n on k vertices in which all edges are coloured blue. That is, the proof is finished

   Similarly, we can reason that when $|B| \geq R(k, \ell - 1)$ then K_n has either an induced subgraph of K_n on ℓ vertices in which all edges are coloured red or an induced subgraph K_n on k vertices in which all edges are coloured blue. Therefore, the proof is finished.

2. The claim can be proved by using induction on $\ell + k$. That is, we firstly prove that the claim holds for the base case $\ell + k = 4$ in which we have to show that $R(2, 2) < \infty$. It is obvious that any 2-coloring on K_2 always results in an induced subgraph of K_2 on 2 vertices in which all edges are coloured either red or blue. Therefore, $R(2, 2) = 2$, and thus the claim holds for the base case. Now, suppose inductively that $R(\ell, k) < \infty$ when $\ell + k = n - 1$, we want to show that $R(\ell, k) < \infty$ when $\ell + k = n$. Recall that, if $\ell, k \geq 2$, we have

   $$R(\ell, k) \leq R(\ell - 1, k) + R(\ell, k - 1)$$

For $\ell + k = n$, both $R(\ell - 1, k)$ and $R(\ell, k - 1)$ must be finite due to the induction hypothesis. It follows from the inequality above that $R(\ell, k)$ must be finite, and thus $R(\ell, k) < \infty$.

□

Problem 2. Let $n \geq 4$ and $\mathcal{A}$ be a set of n-element subsets of $\mathbb{N}$ satisfying $\#\mathcal{A} \leq \frac{4^{n-1}}{3^n}$. Show via a probabilistic argument: There is a map $f : \mathbb{N} \to [4]$ such that $A \in \mathcal{A}$ $\#f(A) = 4$.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of functions $f : \mathbb{N} \to [4]$ and the probability measure $\mathbb{P}$ is given by independent choices in $[4]$ that are uniformly distributed. Consider an arbitrary set $A \in \mathcal{A}$, let Ω_A be the set of all functions $f : \mathbb{N} \to [4]$ such that $\#f(A) = 4$, or $\#f(A) < 4$ since $|[4]| = 4$. This implies that any function $f \in \Omega_A$ must have an element $i \in [4]$ such that $f(a) \neq i$ for any $a \in A$. Consider any element $i \in [4]$, the probability of functions $f \in \Omega$ such that $f(a) \neq i$ for any $a \in A$ is $\left(\frac{3}{4}\right)^n$. Additionally, there are $\binom{4}{1}$ ways to pick an element $i \in [4]$. Therefore,

$$\mathbb{P}(\Omega_A) = \binom{4}{1}\left(\frac{3}{4}\right)^n \tag{2.1}$$

Furthermore, we have

$$\mathbb{P}\left(\bigcup_{A \in \mathcal{A}} \Omega_A\right) \leq \sum_{A \in \mathcal{A}} \mathbb{P}(\Omega_A) \tag{2.2}$$

It follows from $\#A \leq \frac{4^{n-1}}{3^n}$ and (2.1) that

$$\begin{aligned}\sum_{A \in \mathcal{A}} \mathbb{P}(\Omega_A) &< \frac{4^{n-1}}{3^n}\binom{4}{1}\left(\frac{3}{4}\right)^n \\ &< \frac{4^n}{4.3^n}\frac{4!}{3!}\frac{3^n}{4^n} \\ &< 1\end{aligned} \tag{2.3}$$

Due to (2.2) and (2.3), we have

$$\mathbb{P}\left(\bigcup_{A \in \mathcal{A}} \Omega_A\right) < 1$$

This implies that not every function $f \in \Omega$ has the property $\#f(A) < 4$ for at least one $A \in \mathcal{A}$, and thus there must exist a map $f : \mathbb{N} \to [4]$ such that $\#f(A) = 4$ for any $A \in \mathcal{A}$. □

Problem 3. Let $n \geq 2$ and $\mathcal{A}$ be a set of n-element subsets of $\mathbb{N}$ with $\#A < 2^{n-1}$. Show that there is a map $f : \mathbb{N} \to [2]$ such that there is no set $A \in \mathcal{A}$ with $\#f(A) = 1$.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of funtions $f : \mathbb{N} \to [2]$ and the probability measure $\mathbb{P}$ is given by the independent choices in $[2]$ that are uniformly distributed. Consider an arbitrary set $A \in \mathcal{A}$, let Ω_A be the set of all functions $f : \mathbb{N} \to [2]$ such that $\#f(A) = 1$. This implies that any function $f \in \Omega_A$ must have an element $i \in [2]$ such that $f(a) \neq i$ for any $a \in A$. Consider any element $i \in [2]$, the probability of functions $f \in \Omega$ such that $f(a) \neq i$ for any $a \in A$ is $\left(\frac{1}{2}\right)^n$. Additionally, there are $\binom{2}{1}$ ways to pick an element $i \in [2]$. Therefore,

$$\mathbb{P}(\Omega_A) = \binom{2}{1}\left(\frac{1}{2}\right)^n \tag{3.1}$$

Furthermore, we have

$$\mathbb{P}\left(\bigcup_{A\in\mathcal{A}}\Omega_A\right)\leq\sum_{A\in\mathcal{A}}\mathbb{P}(\Omega_A) \tag{3.2}$$

It follows from $\#A<2^{n-1}$ and (3.1) that

$$\begin{aligned}\sum_{A\in\mathcal{A}}\mathbb{P}(\Omega_A)&<2^{n-1}\binom{2}{1}\left(\frac{1}{2}\right)^n\\&<\frac{2^n}{2}\frac{2!}{1!}\frac{1}{2^n}\\&<1\end{aligned} \tag{3.3}$$

Due to (3.2) and (3.3), we have

$$\mathbb{P}\left(\bigcup_{A\in\mathcal{A}}\Omega_A\right)<1$$

This implies that not every function $f\in\Omega$ has the property $\#f(A)=1$ for at least one $A\in\mathcal{A}$, and thus there must exist a function $f:\mathbb{N}\to[2]$ such that $\#f(A)=2$ for any $A\in\mathcal{A}$. Therefore, there must exist a map $f:\mathbb{N}\to[2]$ such that there is no set $A\in\mathcal{A}$ with $\#f(A)=1$. □